Section XIV

NERVOUS SYSTEM DISORDERS

There is no delusion more damaging than to get the idea in your head that you understand the functioning of your brain.

Lewis Thomas
1983

Chapter 45

Disorders of Consciousness

I think, therefore I am.

René Descartes (1644)

The ability to recognize and interact with the surroundings (i.e., consciousness) is the *sine qua non* of the life experience, and loss of this ability is one of the dominant (and dreaded) signs of a life-threatening illness. This chapter describes the major disorders of consciousness that you will encounter in the ICU, with emphasis on delirium, coma, and the ultimate disorder of consciousness, brain death.

ALTERED CONSCIOUSNESS

Consciousness has two components: arousal and awareness.

. *Arousal* is the ability to experience your surroundings, and is also known as *wakefulness*.

. *Awareness* is the ability to understand your relationship to your surroundings, and is also known as *responsiveness*.

These two components can be used to classify the altered states of consciousness, as indicated in Table 45.1 (1–3).

TABLE 45.1 Altered States of Consciousness

Can be Awakened and is Aware	Can be Awakened but is Not Aware	Cannot be Awakened and is Not Aware
Anxiety Lethargy Locked-In State	Delirium Dementia Psychosis Stupor Vegetative State	Coma Brain Death

A L G r a w a n y

Altered States of Consciousness

The principal states of altered consciousness are as follows:

. *Anxiety* and *lethargy* are conditions where arousal and awareness are intact, but there is a change in *attentiveness* (i.e., the degree of awareness).

. A l*ocked-in state* is a condition where arousal and awareness are intact, but there is almost total absence of motor responsiveness. This condition is caused by bilateral injury to the motor pathways in the ventral pons, which disrupts all voluntary movements except up-down ocular movements and eyelid blinking (3).

. *Delirium* and *dementia* are conditions where arousal is intact, but awareness is altered. The change in awareness can be fluctuating (with delirium) or permanent (with dementia).

. *Stupor* is a condition where the subject is difficult to arouse, and is unaware of some or all elements in the environment.

. A *vegetative state* is a condition where there is some degree of arousal (eyes can open), but there is no awareness. Spontaneous movements and motor responses to deep pain can be present, but the movements are purposeless. After one month, this condition is called a *persistent vegetative state* (4).

. *Coma* is the total absence of arousal and awareness (i.e., unarousable unawareness). Like the vegetative state, spontaneous movements and motor responses to deep pain can be present, but the movements are purposeless.

. *Brain death* is similar to coma in that there is a total absence of arousal and awareness. However, brain death differs from coma in two ways: (*a*) it involves loss of all brainstem function, including cranial nerve responses, and spontaneous respirations and (*b*) it is always irreversible.

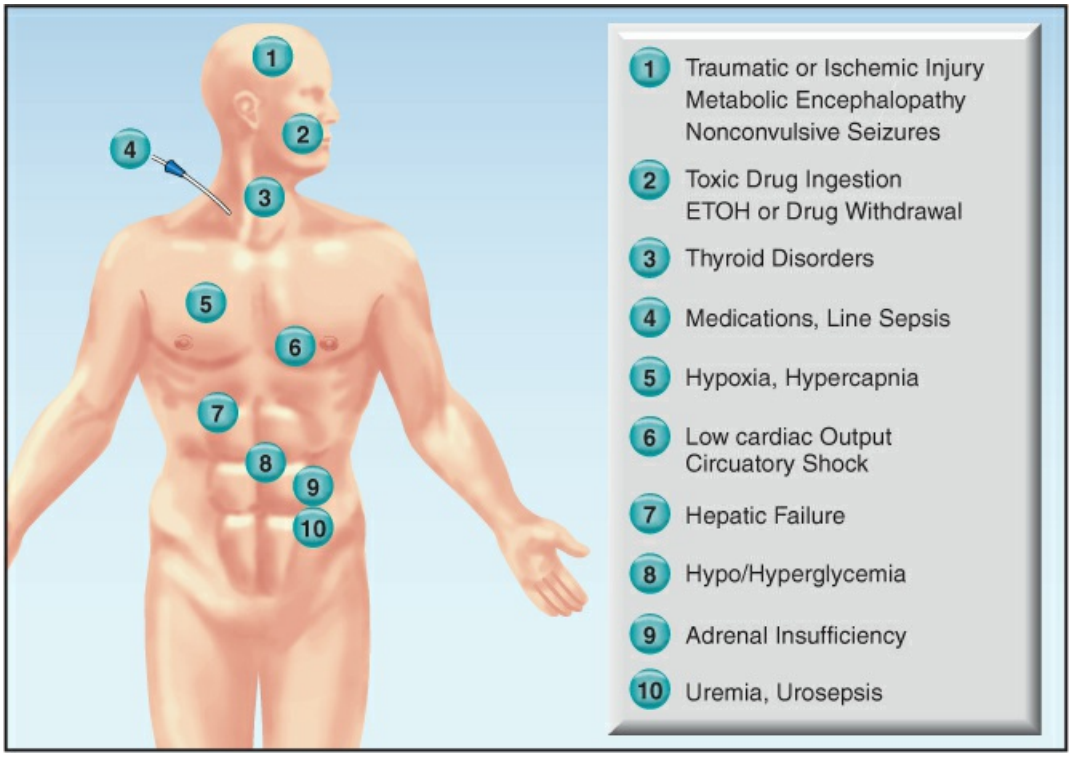


FIGURE 45.1 Common sources of altered consciousness in ICU patients.

Predisposing Conditions

There are an innumerable number of conditions that can alter consciousness, and Figure 45.1 shows a list of the most likely ones in critically ill patients. In a general medical-surgical ICU, the prominent causes of altered consciousness are likely to include the following:

. Sedation during mechanical ventilation.

. Clinical shock syndromes.

. Encephalopathies (ischemic, septic, and metabolic).

. Drug overdoses or withdrawal.

It is worth mentioning that *nonconvulsive seizures* are an often overlooked cause of altered consciousness in ICU patients. (See Chapter 46 for more on this topic.)

DELIRIUM

The delirium described in this section is a manifestation of “brain dysfunction”, which is one part of the multiorgan dysfunction that plagues critically ill patients. The incidence of this type of delirium is about 25–50%, and it is especially common in elderly and ventilator-dependent patients (5). The delirium that is associated with alcohol withdrawal is a separate entity, and is described in a later section.

A L G r a w a n y

Clinical Features

The clinical features of delirium are summarized in Figure 45.2 (6,7). As indicated, delirium is a state of inattentiveness (i.e., inability to focus) that has a rapid onset or a rapidly fluctuating course (i.e., changes occur within a 24-hour time period), and is accompanied by either disordered thinking (i.e., incoherent or illogical responses) or an altered level of consciousness (hyperactive or lethargic).

Three types of ICU-related delirium are recognized: (*a*) *hyperactive delirium*, where the patient is restless and agitated, (*b*) *hypoactive delirium*, where the patient is lethargic and somnolent; and (*c*) *mixed delirium*, where there are alternating episodes of hyperactive and hypoactive delirium. *Hypoactive delirium is the most common form* (5), and is often overlooked because of the popular misperception that delirium is a state of severe agitation.

Predisposing Conditions

Any condition that is associated with an encephalopathy can be a source of delirium. The major ones in the ICU are sepsis, liver failure, uremia, and the use of sedative-hypnotic drugs (5).

SEDATION: The use of heavy and prolonged sedation is considered a risk factor for delirium (8), and *benzodiazepines are considered the principal sedatives that promote delirium in the ICU* (9,10). As a result, avoidance of benzodiazepines is considered an important preventive measure for ICU-related delirium (10).

Diagnosis

Validated screening tools are considered essential for the detection of delirium (5), and the *Confusion Assessment Method for the ICU* (CAM-ICU) is probably the most popular of these diagnostic tools (6,7). A summary of the CAM-ICU method is included in Appendix II at the end of the book.

Delirium versus Dementia

Delirium and dementia are distinct mental disorders that can be confused because they have overlapping clinical features (i.e., attention deficits and disordered thinking). However, *delirium has two characteristics that distinguish it from dementia: i.e., the acute onset and fluctuating course*.

Psychosis versus Delirium

Delirium can be difficult to distinguish from psychosis. Over 40% of hospitalized patients with delirium have psychotic symptoms (e.g., visual hallucinations) (11), and delirium has been given the misnomer of “ICU psychosis” (12).

Management

No drug has proven effective in preventing or correcting ICU-related delirium, and drugs are recommended only to restore calm in severely agitated patients (which is not treating the underlying delirium) (5,10). The drugs that can be used for this purpose are included in Table 45.2. Dexmedetomidine is the preferred agent (10), but can produce troublesome bradycardia and hypotension (see Chapter 6). Haloperidol is a popular agent for hyperactive delirium (see

Chapter 6), but has a relatively slow onset of action (which can be a disadvantage in severely agitated patients). Ziprasidone is an atypical antipsychotic agent that can be only be given intramuscularly, but this can be advantageous when intravenous access is problematic in a severely agitated patient (13).

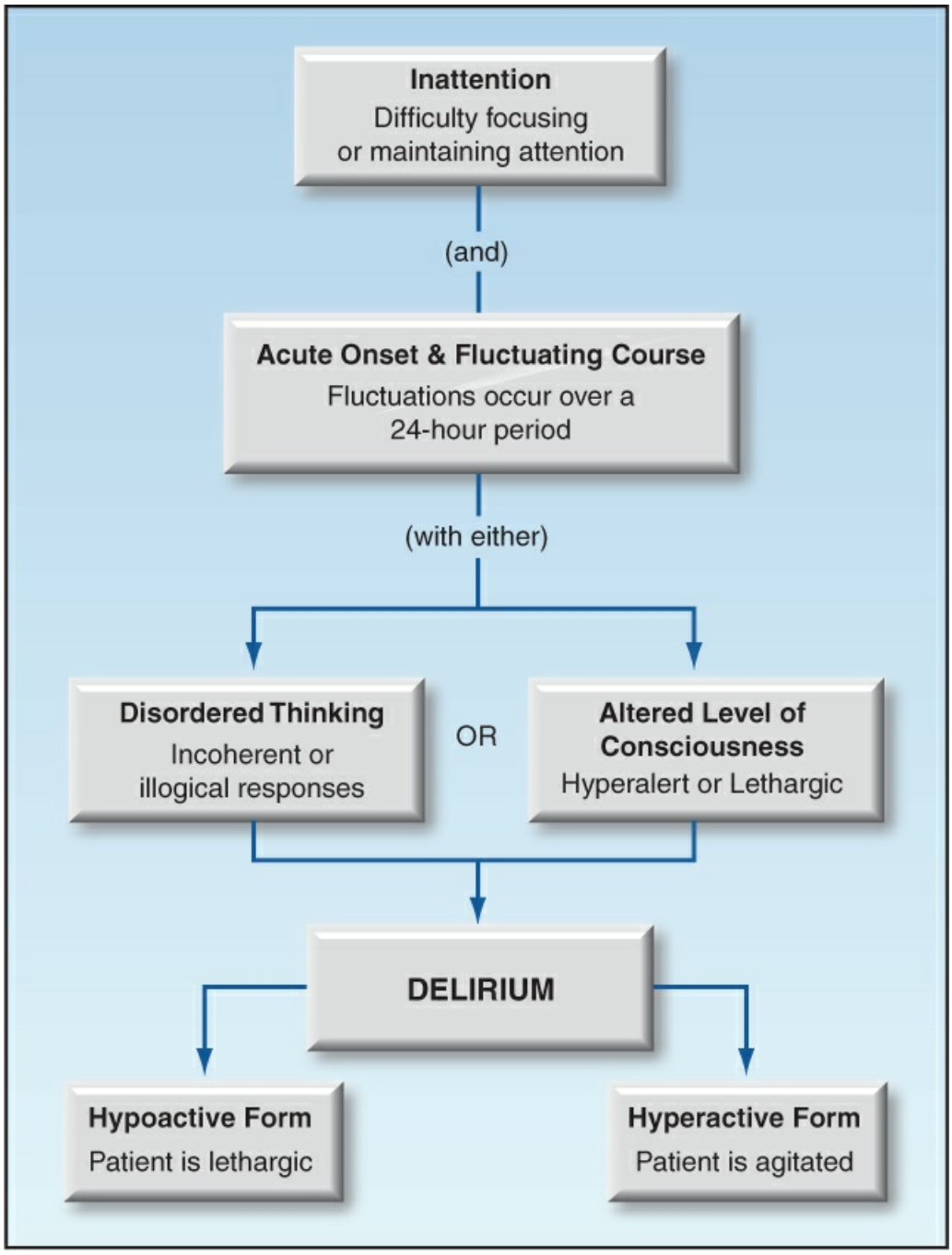


FIGURE 45.2 The clinical features of delirium in critically ill patients.

TABLE 45.2 Drugs for Severe Agitation in ICU-Related Delirium

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Drug	Recommendations
Dexmedetomidine	Dosing: Loading dose (optional) is 1 μg/kg IV over 10 min. followed by an infusion of 0.2–1.5 μg/kg/hr. Comment: Can produce bradycardia and hypotension. A withdrawal syndrome (with hyperexcitability) has been reported.
Haloperidol	Dosing: Start with 5 mg IV as a bolus dose. If no effect after 15 min, double the dose (10 mg IV), or switch to another agent. For maintenance dosing, use 25% of the initial effective dose every 6 hrs. Comment: Has a relatively slow onset of action. Not advised if the corrected QT interval is >500 msec.
Ziprasidone	Dosing: 10 mg IM every 2 hrs, or 20 mg IM every 4 hrs, to a total daily dose of 40 mg. Comment: The IM route can be useful if IV access is problematic.

Prevention

The following interventions are recommended for reducing the risk of ICU-related delirium (5,10): (*a*) adequate treatment of pain, (*b*) avoiding deep sedation, when possible, (*c*) promoting sleep-awake cycles, (*d*) encouraging family visitation, (*e*) early ambulation, and (*f*) avoiding benzodiazepines. Unfortunately, these interventions have not been successful in critically ill patients (14).

ALCOHOL WITHDRAWAL

Alcohol (ethanol) is a central nervous system depressant that acts by stimulating the neuronal release of gamma-aminobutyric acid (GABA), the major inhibitory neurotransmitter in the mammalian brain (15). Chronic alcohol intake increases the density of GABA receptors in the brain (15), and this certainly contributes to the excitatory state that characterizes the alcohol withdrawal syndrome.

Clinical Features

The clinical features of alcohol withdrawal are shown in Table 45.3 (16). The earliest signs of withdrawal are agitation and tremulousness, which can appear as early as 6 hours after the last drink. This can be followed by generalized seizures. hallucinations (visual, auditory, or tactile), and signs of autonomic hyperactivity (e.g., tachycardia, hypertension). Delirium is a relatively late finding, and is accompanied by increased motor activity. The Clinical Institute Withdrawal Assessment of Alcohol Scale, Revised (CIWA-AR) is the standard method for monitoring the severity of alcohol withdrawal, but it is not recommended for patients with alcohol withdrawal delirium (17). Instead, the CAM-ICU assessment of delirium (which is included in Appendix II) is recommended for alcohol withdrawal delirium (17).

Seizures

Generalized seizures can appear as early as 6–8 hours after the last drink (the risk of seizures peaks at 24 hrs), and they can appear in isolation, without other signs of withdrawal (17). The recommended treatment is lorazepam (2 mg IV push), which prevents recurrent seizures (18).

Phenytoin is not recommended for alcohol-withdrawal seizures (17).

Delirium Tremens

About 5% of patients develop a severe form of alcohol withdrawal known as *delirium tremens* (DTs) (16,17), which appears about 2–3 days after the last drink, and is characterized by hyperactive delirium, hallucinations, and signs of autonomic hyperactivity (e.g., fever, tachycardia, hypertension) that can progress to hemodynamic instability. This condition typically lasts 3–5 days (16), and delirium that persists for longer than 5 days should prompt consideration of another cause of the problem (see later). The hypoactive form of delirium, which is a common presentation of ICU-related delirium, does not occur in alcohol withdrawal delirium.

TABLE 45.3 Clinical Features of Alcohol Withdrawal

Features	Onset after Last Drink	Duration
Early Withdrawal Anxiety, Nausea, Tremulousness	6–8 hrs	1–2 days
Generalized Seizures: One or Two	6–48 hrs	2–3 days
Hallucinations: Visual, Auditory, or Tactile	12–48 hrs	1–2 days
Delirium Tremens: Fever, Tachycardia Hypertension, Delirium (Hyperactive), Autonomic Instability	48–96 hrs	3–5 days

Adapted from Reference 16.

Management

The drugs of choice for treating alcohol withdrawal delirium are the benzodiazepines (17), which mimic the CNS-depressant effects of alcohol by stimulating GABA receptors in the brain (15). Large doses of benzodiazepines (e.g., 30–40 mg of IV lorazepam or 150–200 mg of IV diazepam in 3–4 hours) may be required in severe cases (17).

ICU Management

For patients who require ICU-level care, the benzodiazepines should be given intravenously. *Lorazepam* can be given in a dose of 2–4 mg IV every 5–10 minutes until the desired level of calm is achieved. Thereafter, the dose can be reduced to 2–4 mg IV every few hours as needed. Continuous infusions of high-dose lorazepam (i.e., ≥0.1 mg/kg/hr) should not be continued for longer than 48 hours because of the risk of *propylene glycol toxicity* (see Chapter 6). (The same risk occurs with prolonged infusions of diazepam).

It is important to taper benzodiazepines as soon as possible because they are highly lipid soluble and readily accumulate in the brain with continued use, which can delay awakening and prolong the ICU stay. (See Chapter 6 for more on sedation with benzodiazepines.)

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Benzodiazepine Resistance

Benzodiazepine resistance has been described as uncontrolled alcohol withdrawal despite IV lorazepam doses of 30–40 mg or IV diazepam doses of 150–200 mg in the first 3–4 hours of treatment (17). In these cases, the following adjunctive medications have been recommended (17).

. Propofol is a rapidly-acting, GABAergic sedative that is the preferred adjunct to benzodiazepines for patients who are intubated and receiving mechanical ventilation (17). The administration of propofol is described in Chapter 6 (see Table 6.6).

. Phenobarbital is an effective adjunct for benzodiazepines, and can be given in doses of 130–260 mg IV every 15–20 minutes until the desired effect is achieved (19). However, barbiturates should be used cautiously, because they can depress respirations and promote hypotension.

. Dexmedetomidine has been recommended as an adjunct to benzodiazepines (17), using the dosage recommendations in Table 45.2. However, the clinical experience with this drug in severe alcohol withdrawal has not been encouraging (20).

Other Considerations

The following conditions can be mistaken for alcohol withdrawal, and should be considered when there is resistance to benzodiazepines.

Wernicke's Encephalopathy

Patients with alcohol use disorder can experience a sudden deterioration in mental status a few days after admission that is the result of acute Wernicke's encephalopathy (from thiamine deficiency) rather than alcohol withdrawal delirium. This occurs when thiamine stores are borderline depleted on admission, and the glucose load from intravenous fluids and feeding regimens utilizes the remaining thiamine (which is used for glucose metabolism) (21). The presence of nystagmus and oculomotor palsies (e.g., lateral gaze paralysis) can help to identify Wernicke's encephalopathy, but these are infrequent findings.

Patients with alcohol use disorder are typically given thiamine in a dose of 100 mg daily while in the hospital, but this is an insufficient dose for the treatment of Wernicke's encephalopathy. (See Chapter 48 for more on thiamine deficiency.)

Gabapentin Withdrawal

Gabapentin is a GABA analog that is used to treat a variety of disorders, including partial seizures, neuropathic pain, migraine and cluster headaches, psychiatric disorders, and alcohol use disorder. Multiple reports have demonstrated that abrupt discontinuation of gabapentin can result in a hyperactive delirium that is indistinguishable from alcohol withdrawal (22). This usually occurs when the daily dose of gabapentin is 3,000 mg or higher, and the onset can occur from 12 hours to 7 days after the last drug dose. Benzodiazepines are ineffective in controlling the delirium from gabapentin withdrawal, and restarting the gabapentin can have dramatic results (author's experience).

COMA

The patient who is comatose (i.e., unarousable and unaware) is one of the most challenging problems in critical care, as recovery in these patients is never certain.

Etiologies

Coma can be the result of any of the following conditions:

. Diffuse anoxic or ischemic brain injury.

. Supratentorial mass lesion causing transtentorial herniation and brainstem compression.

. Posterior fossa mass lesion causing direct brainstem compression.

. Brainstem stroke.

. Toxic or metabolic encephalopathies (including drug overdose).

. Nonconvulsive status epilepticus.

. Apparent coma (i.e., locked-in state, hysterical reaction).

Bedside Evaluation

The bedside evaluation of coma should include cranial nerve reflexes, spontaneous eye and body movements, and motor reflexes. The following elements of the evaluation deserve mention.

Motor Responses

Spontaneous myoclonus (irregular, jerking movements) can be a nonspecific sign of diffuse cerebral dysfunction, or it can represent seizure activity (myoclonic seizures), while flaccid extremities can indicate diffuse brain injury or injury to the brainstem. A focal motor defect in the extremities (e.g., hemiparesis or asymmetric reflexes) is a sign of a space-occupying lesion or a localized area of brain or spinal cord injury.

RESPONSE TO PAIN: A purposeful response to painful stimulation (e.g., limb withdrawal to pain) is not a feature of the comatose state. Instead, the response to pain is either purposeless or absent. The following responses are determined by a specific area of brain injury:

. With injury to the thalamus, painful stimuli provoke flexion of the upper extremities, which is called *decorticate posturing*.

. With injury to the midbrain and upper pons, the arms and legs extend and pronate in response to pain; this is called *decerebrate posturing*.

. With injury to the lower brainstem, the extremities remain flaccid during painful stimulation.

Eye Opening

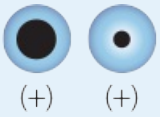
Spontaneous eye opening is an indication of arousal, and is not consistent with the diagnosis of coma. Spontaneous eye opening can be associated with awareness (i.e., locked-in state) or lack of awareness (i.e., vegetative state).

Examination of Pupils

The conditions that affect pupillary size and light reactivity are shown in Table 45.4 (23–25).

A L G r a w a n y

TABLE 45.4 Conditions That Affect Pupillary Size and Reactivity

Pupil Size & Reactivity	Associated Conditions
 (+) (+)	Atropine, anticholinergic toxicity, adrenergic agonists (e.g., dopamine), stimulant drugs (e.g., amphetamines), or nonconvulsive seizures.
 (–) (–)	Diffuse brain injury, hypothermia (<28° C), or brainstem compression from an expanding intracranial mass or intracranial hypertension.
 (–) (+)	Expanding intracranial mass (e.g., uncal herniation), ocular trauma or surgery, or focal seizure.
 (+) (+)	Toxic/metabolic encephalopathy, sedative overdose, or neuromuscular blockade.
 (–) (–)	Acute liver failure, postanoxic encephalopathy, or brain death.
 (+) (+)	Horner's Syndrome
 (+/–) (+/–)	Opiate overdose, toxic/metabolic encephalopathy, hypercapnia, or pontine injury.

(+) and (-) indicate a reactive and nonreactive pupil, respectively. From References 23–25.

Pupillary findings can be summarized as follows:

. Dilated, reactive pupils can be the result of drugs (anticholinergics, CNS stimulants, or adrenergic agonists) or nonconvulsive seizures. Dilated, unreactive pupils can be the result of diffuse brain injury, brainstem compression from an expanding intracranial mass, or increased intracranial pressure.

. A unilateral, dilated and fixed pupil can be the result of ocular trauma or recent ocular surgery, or can be evidence of third cranial nerve dysfunction from an expanding intracranial mass.

. Mid-position, reactive pupils can be the result of a metabolic encephalopathy, a sedative overdose, or neuromuscular blocking drugs, while mid-position, unreactive pupils can be a sign of acute liver failure, postanoxic encephalopathy, or brain death.

. Small, reactive pupils can be the result of a metabolic encephalopathy, while pinpoint pupils can be the result of opioid overdose (pupils reactive) or damage in the pons (pupils unreactive).

Ocular Motility

Spontaneous eye movements (conjugate or disconjugate) are a nonspecific sign of a toxic or metabolic encephalopathy (24), while a fixed gaze preference (one or both eyes) is highly suggestive of a mass lesion or seizure activity.

Ocular Reflexes

The ocular reflexes are used to evaluate the functional integrity of the lower brainstem (24). These reflexes are illustrated in Figure 45.3.

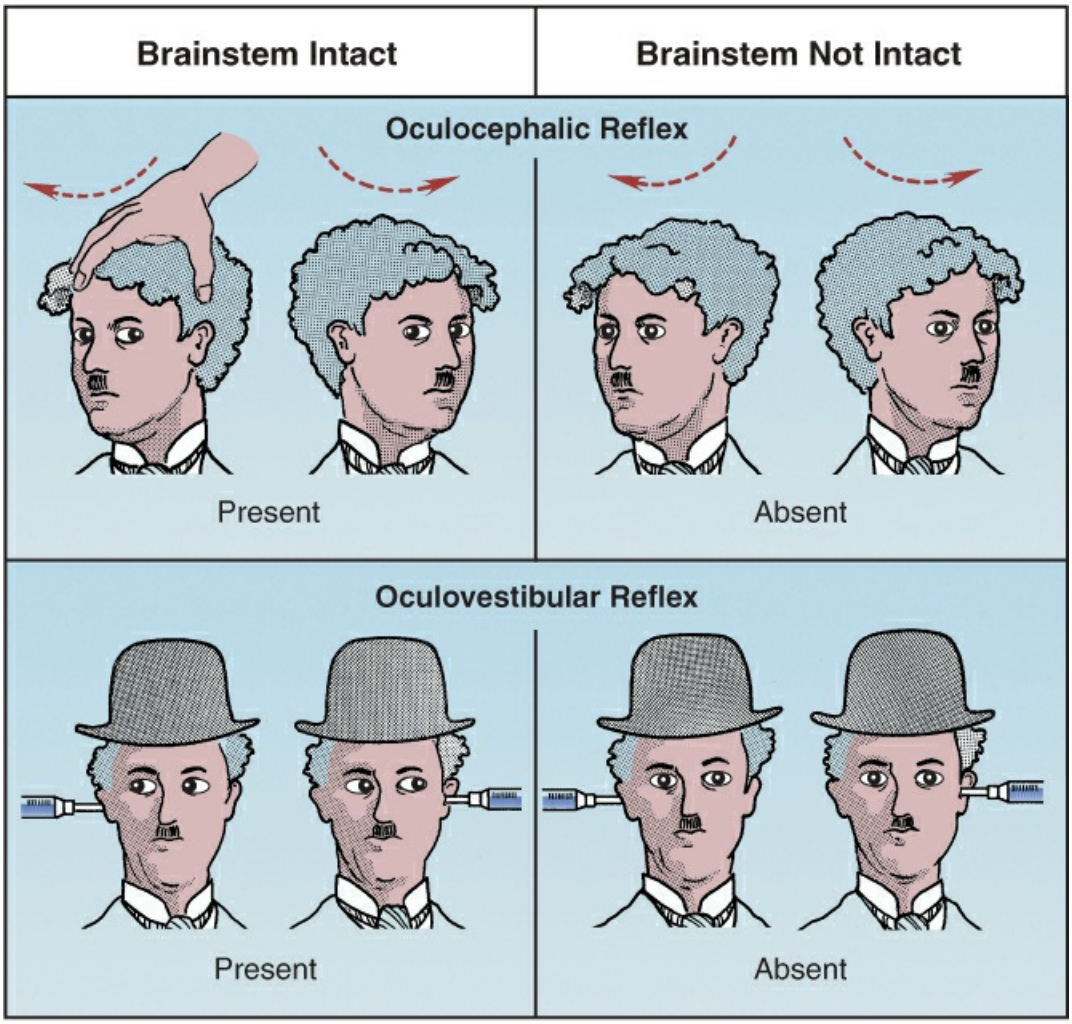


FIGURE 45.3 The ocular reflexes in the evaluation of coma.

OCULOCEPHALIC REFLEX: The oculocephalic reflex is assessed by briskly rotating the head from side-to-side. When the cerebral hemispheres are impaired but the lower brainstem is intact, the eyes will deviate away from the direction of rotation and maintain a forward field of view. When the lower brainstem is damaged (or the patient is awake), the eyes will follow the direction of head rotation. The oculocephalic reflex should *not* be attempted in patients with an unstable cervical spine.

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OCULOVESTIBULAR REFLEX: The oculovestibular reflex is performed by injecting 50 mL of cold saline in the external auditory canal of each ear (using a 50 mL syringe and a 2-inch soft plastic angiocatheter). Before the test is performed, check to make sure that the tympanic membrane is intact and that nothing is obstructing the ear canal. When brainstem function is intact, both eyes will deviate slowly towards the irrigated ear. This conjugate eye movement is lost when the lower brainstem is damaged. After the test is performed on one side, wait 5 minutes before testing the opposite side.

The Glascow Coma Score

The Glasgow Coma Scale, which is shown in Table 45.5, was introduced to evaluate the severity of traumatic brain injuries (26,27), but has since been adopted for use in nontraumatic brain injuries (28). The Scale consists of three components: 1) eye opening, 2) verbal communication, and 3) motor response to verbal or noxious stimulation. The *Glasgow Coma Score* (GCS) is the sum of the three components. A minimum score of 3 indicates total absence of awareness and responsiveness, while a maximum score of 15 is normal.

TABLE 45.5 The Glasgow Coma Scale and Score

	Points	
Eye Opening:		
Spontaneous	4	
To speech	3	
To pain	2	
None	1	☐ Points
Verbal Communication:		
Oriented	5	
Confused conversation	4	
Inappropriate but recognizable words	3	
Incomprehensible sounds	2	
None	1	☐ Points
Motor Response:		
Obeys commands	6	
Localizes to pain	5	
Withdraws to pain	4	
Abnormal flexion (decorticate response)	3	
Abnormal extension (decerebrate response)	2	
None	1	☐ Points
Glasgow Coma Score (Total of 3 Scales)*		☐ Points

*Worst score is 3 points, and best score is 15 points. With endotracheal intubation, the highest score is 11.

Interpretations

The GCS is not reliable in patients who are paralyzed, heavily sedated, or hypotensive.

Otherwise, the GCS (best score) can be used as follows:

. To define coma (GCS ≤8).

. To stratify the severity of head injury (mild: GCS = 13–15, moderate: GCS = 9–12, severe: GCS ≤8) (26,27).

. To identify candidates for intubation; i.e., airway protective reflexes are typically defective at a GCS ≤8, which is used as an indication for endotracheal intubation.

. As a prognostic marker; e.g., in the initial evaluation of nontraumatic coma, patients with a GCS ≤6 are seven-times more likely to awaken than patients with a GCS ≤5 (27).

INTUBATED PATIENTS: One of the major shortcomings of the Glascow Coma Scale is the inability of evaluate verbal responses in intubated patients. These patients are assigned a verbal "pseudoscore" of 1 (for a maximum GCS of 11).

BRAIN DEATH

The Uniform Determination of Death Act states that "An individual who has sustained either (*a*) irreversible cessation of circulatory and respiratory functions, or (*b*) irreversible cessation of all functions of the entire brain, including the brainstem, is dead" (29). The evaluation of these requirements is the purpose of the brain death determination.

Brain death is not a common consequence of the nontraumatic coma that is typically encountered in the ICU. It is most often the result of traumatic brain injury or intracerebral hemorrhage, where increases in intracranial pressure are sufficient to cause complete cessation of cerebral blood flow (30). Suspicion of brain death is often prompted by the absence of pupillary light reflexes, or the lack of spontaneous breathing efforts during mechanical ventilation.

Prerequisite Conditions

A checklist for the brain death determination in adults is shown in Table 45.6 (30–32). This checklist begins with conditions that must be satisfied before the brain death determination is performed. The following are some points of interest about the prerequisite conditions.

. Although a systolic pressure ≥100 mm Hg is included as one of the conditions, the mean arterial pressure is the more important physiological pressure, and should be measured if an arterial catheter is in place. Note that the required mean arterial pressure (≥75 mm Hg) is higher than usual target for treating hypotension (≥65 mm Hg). This will help to counteract the effect of increased intracranial pressure to impede cerebral perfusion.

. The absence of effects from CNS-suppressant drugs can be difficult to determine if sedative drugs were used recently. When blood levels of a sedating drug are not available, waiting 5 half-lives since the last dose is recommended to ensure drug clearance from the bloodstream (although this does not guarantee clearance from the brain) (31).

. If targeted temperature management was used with a target below 36° C, then a body temperature of >36° C should be maintained for at least 24 hours before the brain death determination (31).

A L G r a w a n y

When the prerequisite conditions are satisfied, the brain death determination can begin. This includes two assessments that are separated by at least 12 hours. One examiner is considered sufficient (in adults) if the examiner is experienced in the task.

TABLE 45.6 Checklist for Brain Death Determination in Adults

Instructions: The patient can be declared legally dead if Steps 1–4 are cofirmed, or there is a positive confirmatory test.	**Check (√) Item if Confirmed**	
Step 1: Prerequisites: All of the following conditions should be satisfied prior to the brain death determination. • At least 48 hrs has elapsed since the brain injury. • The systolic pressure is ≥100 mm Hg, or the mean arterial pressure is ≥75 mm Hg. • The body temperature is >36° C (96.8° F). • There is no hypothyroidism or hypoglycemia. • There are no CNS-depressant drug effects.	☐	
Step 2: Establish the Cause of Coma: The cause of coma is known, and is sufficient to cause irreversible brain death.	☐	
Step 3: Absence of Brain, and Brainstem, Function: Two examinations are required, separated by at least 12 hrs. A single examiner is sufficient, if adequately trained. A. The patient is comatose (unaware and unresponsive). B. The following brainstem reflexes are absent: • Absent pupillary response to bright light. • Absent corneal reflex. • Absent gag and cough reflexes • Absent oculocephalic reflex • Absent oculovestibular reflexes (bilaterally)	**First Exam** ☐ ☐	**Second Exam** ☐ ☐
Step 4: Absence of Spontaneous Breathing Efforts: There are no spontaneous breathing efforts when the arterial PCO_2 is >60 mm Hg, and >20 mm Hg above the patient's baseline level.	☐	
Step 5: Confirmatory Tests: Confirmatory tests are indicated only when Steps 1–4 cannot be completed or unequivocally interpreted. Acceptable tests are (1) transcranial Doppler ultrasound, (2) radionuclide cerebral blood flow scan, or (3) 4-vessel catheter angiography.		

Adapted from the clinical practice guideline in Reference 31.

Brainstem Reflexes

The brain death determination begins with an evaluation of the reflexes that require a functioning brainstem: i.e., the corneal and pupillary light reflexes, the oculocephalic and oculovestibular reflexes, and the gag and cough reflexes. All these reflexes must be absent to proceed with the evaluation.

The Apnea Test

The most convincing evidence of brain death is the absence of spontaneous breathing efforts in the face of an increased arterial PCO_2 ($PaCO_2$). (*Note:* CO_2 is a potent respiratory stimulant: i.e.,

an increase in $PaCO_2$ of only 1 mm Hg will increase the minute ventilation by 2 L/min (32).) The evaluation of spontaneous breathing efforts is called the *apnea test*, and it proceeds as follows:

. Prior to the test, the patient is preoxygenated with 100% O_2, and an arterial blood gas is obtained to establish the baseline $PaCO_2$.

. The patient is then separated from the ventilator and oxygen is insufflated into the endotracheal tube (apneic oxygenation) to help prevent O_2 desaturation during the apneic period.

. The goal of the apnea test is to allow the $PaCO_2$ to rise 20 mm Hg above baseline. The $PaCO_2$ rises about 3 mm Hg per minute during apnea at normal body temperatures (33), so an apnea period of 6–7 minutes should be sufficient for reaching the target $PaCO_2$. A repeat arterial blood gas is obtained at the end of the apnea period, and the patient is placed back on the ventilator.

. If apnea persists despite a rise in $PaCO_2$ ≥20 mm Hg, the test confirms the diagnosis of brain death.

. If the second apnea test confirms the diagnosis of brain death, *the time of death is the time that the second blood gas results became available* (31). However, patients are often kept on mechanical ventilation for a short period of time after this, to allow family members to have a final visit with the patient.

The apnea test is risky, because O_2 desaturation, hypotension, and serious cardiac arrhythmias can appear during the apnea period (34). If the apnea test cannot be completed, confirmatory testing is required to establish the diagnosis of brain death.

Lazarus' Sign

Brain-dead patients can exhibit brief, spontaneous movements of the head, torso, or upper extremities (*Lazarus' Sign*), especially after they are removed from the ventilator (35). These movements are the result of neuronal discharges in the cervical spinal cord, possibly in response to hypoxemia, and they can be a source of angst when they appear after the patient has been pronounced brain dead and is removed from the ventilator.

Confirmatory Tests

The acceptable confirmatory tests are included in Table 45.6. Confirmatory tests that are not considered acceptable include electroencephalography (EEG), auditory and somatosensory evoked potentials, computed tomographic angiography, and magnetic resonance angiography (31).

A FINAL WORD

Family Care

In the care of the patient with persistent coma or a persistent vegetative state, spending time with the patient's family and/or close acquaintances to explain what is happening (and what is likely

to happen), is as vital as patient care. These people will look to you for guidance in making end-of-life decisions, and avoiding *the conspiracy of silence* (36) is one of the greatest services you can perform as a physician.

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Altered Consciousness

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A L G r a w a n y

Chapter 46

Disorders of Movement

When we contemplate this life, we see motion as its principal characteristic, and when we take a farther view, we see that this motion must necessarily waste the machine in which it resides.

John Young (*a*)

This chapter describes three movement disorders that you are likely to encounter in the ICU: (*a*) involuntary movements (i.e., seizures), (*b*) weak or ineffective movements (i.e., neuromuscular weakness), and (*c*) no movements (i.e., drug-induced paralysis). These conditions are aberrations of the motion described by John Young in the introductory quote, but they retain the ability to "waste" the human machine.

SEIZURES

The risk of seizures in critically ill patients is determined primarily by the presence and type of brain injury, rather than the general condition of the patient (e.g., severity of shock). The conditions that typically produce new-onset seizures are listed in Table 46.1, along with the reported frequency of seizures (1–3). Also included in Table 46.1 is the frequency of troublesome seizures that either persist or recur before the patient recovers from the prior one. This condition is known as *status epilepticus*, and it is the principal focus of this section.

TABLE 46.1 Reported Occurrence Rates of Seizures

Condition	Seizures	Status Epilepticus
Traumatic Brain Injury	20–50%	15–25%
Ischemic Brain Injury	40%	30%
Encephalitis*	35–45%	20%
Intracerebral Hemorrhage	20–30%	10–20%
Subarachnoid Hemorrhage	10–20%	15%

No Neurologic Illness	4–15%	<1%
Acute Ischemic Stroke†	2–5%	—

*Includes infectious and autoimmune encephalitis.
† Includes only the early period (one week) after acute stroke.
Data from References 1–3.

Types of Seizures

Seizures are classified by the presence or absence of abnormal movements (convulsive versus nonconvulsive seizures), the type of abnormal movement (e.g., tonic–clonic, myoclonic), and the extent of involvement (generalized versus focal seizures).

Abnormal Movements

The movements associated with seizures can be *tonic* (caused by sustained muscle contraction), *clonic* (rhythmic movements with a regular amplitude and frequency) or *myoclonic* (irregular, jerky movements that vary in amplitude and frequency). The abnormal movements can also be familiar but repetitive (e.g., chewing or lip smacking); these are called *automatisms*. Of interest, myoclonus is not universally regarded as a seizure because it is not associated with rhythmic discharges on the EEG (4).

Generalized vs. Focal Seizures

Generalized seizures arise from synchronous, rhythmic electrical discharges that involve most of the cerebral cortex, and they are always associated with loss of consciousness. However, they are not always associated with abnormal (convulsive) movements. Focal seizures arise from discrete areas of the brain, and may or may not be accompanied by loss of consciousness. However, focal seizures often progress to generalized seizures (1).

Partial Complex Seizures

Partial complex seizures are nonconvulsive seizures that produce behavioral changes. The typical manifestation is a patient who is awake but not aware of the surroundings (similar to absence seizures). These are often preceded by an *aura* (e.g., a particular smell), and can be accompanied by automatisms.

Status Epilepticus

Status epilepticus (SE) in adults is traditionally defined as *≥5 minutes of continuous seizure activity, or two or more discrete seizures without an intervening period of consciousness* (5). There are about 20 different types of SE (5), but the ones most likely to be seen in the ICU are (*a*) generalized convulsive and nonconvulsive SE, (*b*) myoclonic SE, and (*c*) focal motor SE (with and without loss of consciousness). Nonconvulsive SE is of particular interest as a source of impaired consciousness.

TABLE 46.2 Clinical Manifestations of Nonconvulsive Status Epilepticus

Impaired Consciousness	Others

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Confusion (49%) Coma (22%) Lethargy (21%)	Speech Disturbances (15%) Myoclonus (13%) Bizarre Behavior (11%) Agitation or Delirium (8%) Hallucinations (6%)

Frequency of occurrence in parentheses. From Reference 6.

Nonconvulsive SE

Nonconvulsive SE is an elusive diagnosis because the clinical manifestations are varied and atypical for seizures. This is demonstrated in Table 46.2, which shows the clinical manifestations in 105 reported cases of nonconvulsive SE (6). Impaired consciousness (i.e., confusion, coma, or lethargy) was the most common manifestation, and was present in about 80% of cases. The remaining manifestations included speech disturbances, bizarre behavior, agitation, delirium, and hallucinations.

The importance of nonconvulsive seizures as a cause of impaired consciousness is demonstrated by the following observations:

. In ICU patients with impaired consciousness, nonconvulsive seizures have been detected in 16% of the patients when continuous EEG monitoring is used (usually for 1–2 days) (7), and in 4–8% of the patients when intermittent EEG recordings are used (7,8).

. Nonconvulsive seizures have been detected in 37% of the EEGs obtained for the evaluation of impaired consciousness (9).

Observations like these emphasize the need to consider nonconvulsive seizures in any ICU patient with impaired consciousness, and to use continuous EEG monitoring when available.

Treatment

The pharmacotherapy of SE is divided into three steps, and the drugs involved in each of these steps are presented in Tables 46.3 and 46.4. Seizures that continue uninterrupted for longer than 30 minutes can result in permanent neuronal injury (5), emphasizing the need for prompt suppression of SE. *Note:* The treatment of SE presented here is primarily intended for generalized convulsive SE (10), but it is also used (with less success) for nonconvulsive SE.

Step 1: Benzodiazepines

Benzodiazepines are the first-line drugs of choice for the rapid termination of generalized SE (10). *Intravenous lorazepam is the preferred agent* (10,11), *and intramuscular midazolam is effective when IV access is problematic* (12). Intravenous diazepam is also effective, but is not favored because of rapid washout from the brain, which increases the risk of seizure recurrence (1).

LORAZEPAM: Intravenous lorazepam can terminate generalized convulsive SE in about two minutes, and the effect lasts for 6–12 hours (11). The recommended dosage is 0.05–0.1 mg/kg (or 4 mg) IV over 2 minutes, which can be repeated once if needed.

MIDAZOLAM: Midazolam can be given by IV or intramuscular (IM) injection, and the

recommended dose is 0.15 mg/kg, or 10 mg (10). IM midazolam is as effective as IV lorazepam (12), and the IM route is advantageous when IV access is problematic (e.g., in the prehospital setting).

Step 2: Anticonvulsants

About one-third of cases of generalized SE do not respond to benzodiazepines (13), and the next step is the administration of a nonsedating anticonvulsant: i.e., levetiracetam, fosphenytoin, or valproic acid. All are equivalent in terminating SE (which occurs in about half of the cases) (14), but *levetiracetam is often favored because it has fewer side effects.*

LEVETIRACETAM (KEPPRA®): The recommended dose of levetiracetam in generalized SE is 60 mg/kg IV, infused over 5–15 minutes, with a maximum single dose of 4,500 mg (10). If necessary, the drug can be continued at a dose of 1,000 mg IV every 12 hours, and serum drug levels can be used to adjust dosage. There is no risk of hypotension or respiratory depression, and no drug interactions.

FOSPHENYTOIN: Fosphenytoin is a water-soluble prodrug that produces less cardiac depression and hypotension than phenytoin because it does not contain the solvent propylene glycol. (See Chapter 6 for a description of propylene glycol toxicity.) As a result, fosphenytoin can be infused three times faster than phenytoin (at 150 mg/min) (15). The dosing is the same as phenytoin, but is expressed as phenytoin equivalents (PE), as shown in Table 46.3. If the drug is effective, maintenance therapy is started at 100 mg PE IV every 8 hours, and serum levels are monitored to guide dosing.

The parent drug, phenytoin, has several undesirable features, including CNS toxicity (lethargy, confusion, ataxia) and multiple drug interactions (due to its metabolism by the cytochrome P450 system in the liver). The conversion of fosphenytoin to phenytoin is increased in liver failure, and careful monitoring of serum phenytoin levels is advised in patients with hepatic insufficiency.

TABLE 46.3 Drugs Regimens for Status Epilepticus

Drug	Recommendations
Step 1: Benzodiazepines	
Lorazepam	Dosing: 0.1 mg/kg (or 4 mg) IV over 2 min. Repeat in 5 min, if needed. Comment: The preferred drug regimen for terminating SE. Onset of action is <2 min, and effect lasts 6–12 hrs.
Midazolam	Dosing: 0.15 mg/kg (or 10 mg) IV, or by intramuscular (IM) injection. Comment: IM midazolam is as effective as IV lorazepam, and the IM route is advantageous when IV access is problematic (e.g., in the field).
Step 2: Anticonvulsants	
Levetiracetam	Dosing: 60 mg/kg IV over 5–10 min. Max single dose is 4,500 mg. Comment: Is often preferred to the other anticonvulsants because it has few side effects.

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Fosphenytoin	Dosing: 20 mg PE/kg IV at ≤150 mg/min. Max single dose is 1,500 mg PE. Comment: A water soluble prodrug with less risk of hypotension than phenytoin because it does not contain the solvent propylene glycol.
Valproic Acid	Dosing: 40 mg/kg IV over 5–10 min. Max single dose is 3,000 mg. Comment: As effective as the other anticonvulsants. Only side effect of con-cern is hyperammonemia, which creates a risk of encephalopathy.

Drug dosing from the guidelines in Reference 10. PE = phenytoin equivalents.

VALPROIC ACID: Valproic acid is equivalent to the other anticonvulsants for terminating SE (14) (see Table 46.3 for the recommended dose), and has relatively few side effects (16). However, it is far less popular than the other anticonvulsants (author's observation). One side effect that has gained some attention is a 40% risk of hyperammonemia (17), which can produce an encephalopathy.

Step 3: Refractory Status Epilepticus

About 10–15% of cases of generalized SE are refractory to step 1 and 2 drugs, and the treatment for these cases is anesthetic doses of one of the drugs in Table 46.4. There is no evidence that one drug regimen is superior to the others, and selection is often based on familiarity. Patients should be intubated and on mechanical ventilation prior to starting these drugs, as they all depress ventilation in the recommended doses. Continuous EEG monitoring is also advised, and at this point, consultation with a critical care neurologist or epileptologist is the best option. (*Note:* Drugs other than those in Table 46.4 can be used for refractory SE, including thiopental, phenobarbital, and ketamine. The dosing regimens for these drugs are available in Reference 1.)

Outcomes

The in-hospital mortality rates are as high as 21% for generalized convulsive SE, about 50% for generalized nonconvulsive SE, and as high as 61% for refractory SE (7).

TABLE 46.4 Drug Regimens for Refractory Status Epilepticus

Drug	Dosing Regimens
Propofol	Start with an IV bolus dose of 1–2 mg/kg, and begin infusion at 1 mg/kg/hr. Titrate upward as needed to max dose rate of 15 mg/kg/hr (or 5 mg/kg/hr if infusion longer than 48 hrs).
Midazolam	Load with 0.2 mg/kg IV, then infuse at 0.2 mg/kg/hr, and titrate upward as needed to a max. dose rate of 4 mg/kg/hr.
Pentobarbital	Load with 5–15 mg/kg IV over one hr, then begin infusion at 1 mg/kg/hr. Titrate upward, if needed, to a max. dose rate of 5 mg/kg/hr.

Dosing regimens from Reference 1.

NEUROMUSCULAR WEAKNESS SYNDROMES

The following is a description of neuromuscular weakness syndromes that you may encounter in the ICU.

Myasthenia Gravis

Myasthenia gravis (MG) is an autoimmune disease produced by antibody-mediated destruction of acetylcholine receptors on the postsynaptic side of neuromuscular junctions (18). Skeletal muscles are the only target of this illness.

Predisposing Conditions

MG can be triggered by major surgery or a concurrent illness. Thymomas are responsible for 10% of cases of MG (18). MG can also be aggravated by major surgery, physiological stress, and drugs that can impair neuromuscular transmission. The drugs that can aggravate MG are listed in Table 46.5. These drugs should not be used in patients with MG. However, note that high-dose corticosteroids are listed as aggravating MG, which is concerning because high-dose steroids are used to *treat* MG. (For more on this, see Reference 20.)

TABLE 46.5 Drugs that can Exacerbate Myasthenia Gravis

1. *Antibiotics:* aminoglycosides, fluoroquinolones, macrolides, penicillins
2. *Cardiovascular Drugs:* β-blockers, calcium channel blockers, class Ia antiarrhythmics (procainamide, quinidine, disopyramide)
3. *Neuromuscular Blockers:* both depolarizing and non-depolarizing agents
4. *Others:* high-dose steroids*, inhalational anesthetics, magnesium, lithium

*High-dose steroids are also used to treat myasthenia gravis. The list of drugs is from References 19 and 20.

Clinical Features

Skeletal muscle weakness is the sole manifestation of MG, and the weakness has the following characteristics:

. The eye muscles (i.e., eyelid and oculomotor muscles) are always involved, and the earliest signs of MG are diplopia and ptosis. Symmetrical limb weakness usually follows, but in 15% of cases, the weakness is confined to the eye muscles (18).

. The muscle weakness typically worsens with muscle activity, and improves with rest.

. The deficit is purely motor, and deep tendon reflexes are preserved.

. Progressive disease leads to weakness of the diaphragm and chest wall muscles, and rapid progression to respiratory failure, called *myasthenic crisis,* occurs in 15–20% of patients (21).

Diagnosis

The diagnosis of MG is suggested by the pattern of muscle weakness, with ptosis as a dominant feature. Confirmatory diagnostic tests include the ice test, and serum antibody levels.

ICE TEST: The ice test is based on the premise that cooling can improve neuromuscular transmission in MG. The test is performed on an eyelid with ptosis, and the distance between the upper and lower eyelids (the edges of the palpebral fissure) is measured. An ice pack (or plastic glove filled with ice) is then placed on the eyelid (with the eyes closed) for 2 minutes. An increase of 2 mm in the distance between the upper and lower eyelids is a positive test, and helps

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confirm the diagnosis of MG. This is illustrated in Figure 46.1. A positive ice test has a sensitivity and specificity of 80% for the diagnosis of MG (22).

AUTOANTIBODY ASSAYS: The presence of acetylcholine (Ach) receptor antibodies in the serum confirms the diagnosis of MG. About 10–15% of patients with MG will not have Ach antibodies in the blood (18), and for these cases, a muscle-specific kinase (MuSK) antibody should be assayed. This antibody is present in about 40% of MG patients who have no detectable Ach antibodies (23).

Management in the ICU

Most cases of MG that are severe enough to warrant ICU-level care have respiratory muscle weakness, with some degree of respiratory insufficiency. The principal concern in these patients is determining when intubation is necessary.

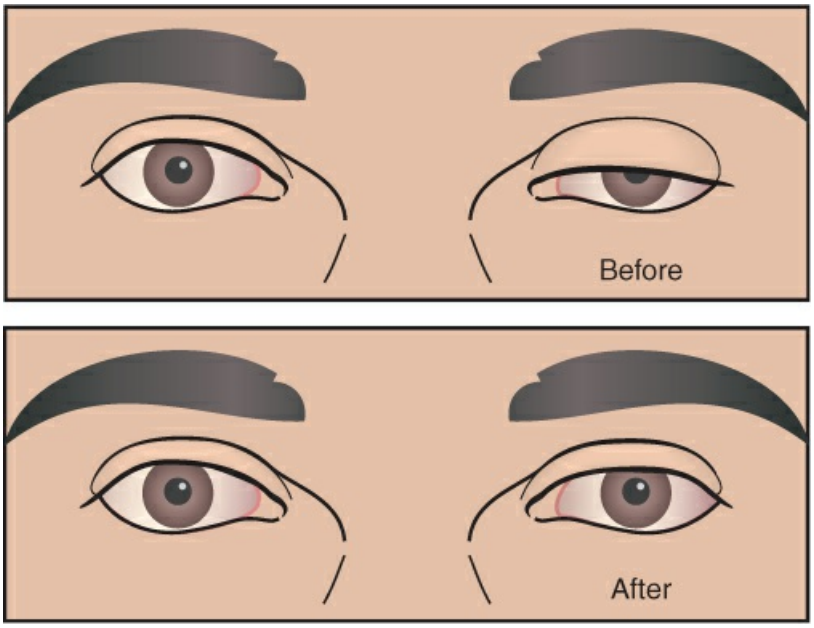


FIGURE 46.1 Illustration of a positive ice test for the diagnosis of myasthenia gravis. An icepack is applied to the eyelid with ptosis for 2 minutes, and this results in an increase in the distance between the upper and lower eyelids.

INTUBATION: The deterioration in ventilatory status can be very rapid in MG, so intubation should not be delayed until the patient is *in extremis*. Measurements of vital capacity and maximum inspiratory pressure (i.e., the negative pressure generated by a maximum inspiratory effort against a closed valve) can be used to guide decisions about intubation. *A vital capacity of ≤15 mL/kg (or <1 liter), and a maximum inspiratory pressure of –20 cm H_2O (normal being >–100 cm H_2O) should prompt serious consideration of intubation and mechanical ventilation* (21). Noninvasive ventilation with BIPAP (described in Chapter 26) should only be considered if an improvement in respiratory status is expected in the next few days (which is usually not the case).

TREATMENT FOR MYASTHENIC CRISIS: There are two options for treating severe MG with

respiratory failure (i.e., myasthenic crisis) (21):

. *Intravenous immunoglobulin*, or IVIG, (to neutralize the pathologic antibodies) in a dose of 2 grams per kilogram body weight, administered over 2–5 days.
. *Plasmapheresis* (to remove the pathologic antibodies from the bloodstream), which typically involves 5 sessions over 7–10 days, with each session exchanging 1.0 to 1.5 plasma volumes.

Both treatments are equally effective, and will produce a favorable response in about 70% of patients (21). However, plasmapheresis is favored because it produces a faster response: i.e., the response to IVIG is not evident for 4–5 days, while the response to plasmapheresis may be apparent after the second session (21).

PYRIDOSTIGMINE: Pyridostigmine (Mestinon®) is an acetylcholinesterase inhibitor that is typically given to patients with mild-to-moderate MG (24,25). However, the drug is not used when MG progresses to respiratory failure, because one of the side effects is to increase respiratory secretions (cholinergic effect).

CORTICOSTEROIDS: High-dose methylprednisolone (2 grams daily for two days) has been used to treat myasthenic crisis, with some evidence of success (26). However, this treatment is debated because of reports that high-dose corticosteroids can also aggravate MG, as mentioned earlier. It certainly seems wise to withhold this treatment in patients who are responding satisfactorily to IVIG or plasmapheresis.

OTHER CONSIDERATIONS:

. Prophylaxis for venous thromboembolism (e.g., enoxaparin 40 mg daily) is considered particularly important in diseases with neuromuscular weakness, because of limited mobility.
. Patients with bulbar weakness can also have difficulty swallowing, which creates a risk of aspiration of saliva into the airways. Patients who are not intubated should therefore have a swallowing evaluation to assess the risk of aspiration.
. Patient's with new-onset MG should have a contrast-enhanced CT of the chest to search for a thymoma. If one is present, a thymectomy may be appropriate to reduce the need for immunotherapy. This decision, and the use of immunotherapy for long-term management of MG, will be the purview of a neurologist.

Outcomes

About 95% of patients who are hospitalized with MG will survive the hospitalization (21), but few will recover completely (18).

Guillain–Barré Syndrome

The Guillain–Barré syndrome (GBS) is an *inflammatory demyelinating polyneuropathy* that often follows an acute infectious illness (by 1 to 3 weeks). Infectious organisms that predispose to GBS include *Campylobacter jejuni*, Zika virus, and SARS coronavirus-2 (27). An immune mechanism is suspected, and *molecular mimicry* has been implicated (27). (*Note:* In molecular

mimicry, antibodies to a component of pathogens cross-react with a component of peripheral nerves.)

Clinical Features

The clinical presentation of GBS is characterized by paresthesias and symmetrical limb weakness (upper and lower limbs) that progresses over a few days to a few weeks. Progression to respiratory failure occurs in 25% of cases (28), and autonomic instability is a feature in advanced cases (29). The condition resolves spontaneously in about 80% of cases, but residual neurological deficits are common (28).

The features that distinguish GBS from myasthenia gravis are shown in Table 46.6. There is no oculomotor weakness in GBS, although there is a GBS variant known as the *Miller Fisher syndrome* that presents with ophthalmoplegia and ataxia (30). Another characteristic feature of GBS is the deep tendon reflexes, which are diminished or absent.

Diagnosis

The diagnosis of GBS is based on the clinical presentation (symmetrical limb weakness presenting within a few weeks of an infection), and nerve conduction studies that show a decrease in nerve conduction velocity. Lumbar puncture is often performed to rule out other conditions, and the cerebrospinal fluid shows elevated protein levels in patients with GBS (31).

TABLE 46.6 Features that Distinguish Guillain–Barré Syndrome from Myasthenia Gravis

Feature	Guillain–Barré Syndrome	Myasthenia Gravis
Oculomotor Weakness	No	Yes
Fatigable Weakness	No	Yes
Deep Tendon Reflexes	Diminished	Intact
Autonomic Instability	Yes	No
Nerve Conduction	Slowed	Normal
Cerebrospinal Fluid	↑ Protein	Normal

Treatment

Patients with GBS who require ICU-level care are managed in a very similar fashion to the management described for myasthenia gravis.

. Respiratory deterioration can be rapid, and monitoring the vital capacity and negative inspiratory pressure can help in timely intubation, before the patient is *in extremis*.

. Patients who are unable to walk unaided, or have respiratory failure, should be treated with *intravenous immunoglobulin or plasmapheresis* (31), using the same regimens described earlier for myasthenia gravis. Both treatments are considered equivalent, and no preference is given in the most recent guidelines on GBS (31).

PAIN: Pain can be prominent in the acute and resolving stages of GBS, and is considered to be neuropathic rather than nociceptive pain. The recommendation from the guidelines on GBS is *gabapentin* (Neurontin®) in a dose of 300 mg three times daily (31). However, opioids should also be considered for more immediate pain relief, since they can be effective for relieving neuropathic pain (32).

Outcomes

In the United States, about 97% of patients who are hospitalized for GBS will survive the hospital stay (33), but 10–15% of these patients will be severely disabled (34).

Critical Illness Neuromyopathy

The disorders known as *critical illness polyneuropathy* (CIP) and *critical illness myopathy* (CIM) are secondary disorders, and typically occur in patients with severe systemic inflammatory diseases (35–37). These disorders often co-exist in the same patient, and become apparent when patients fail to wean from mechanical ventilation.

Pathogenesis

. CIP is a diffuse sensory and motor axonal neuropathy that is particularly prevalent in patients with severe sepsis and septic shock (37). It affects both limb and truncal muscles, and is considered the most common peripheral neuropathy in critically ill patients (38).

. CIM is a diffuse inflammatory myopathy that involves both limb and truncal muscles (39). Predisposing conditions include severe sepsis and septic shock, immobility (e.g., by neuromuscular paralysis), and hyperglycemia (36,37). An acute myopathy that resembles CIM has also been reported in ventilator-dependent asthmatic patients who were paralyzed and treated with high dose steroids (40).

The common denominator in both CIP and CIM is severe or progressive systemic inflammation (36,37), so both conditions could represent inflammatory organ injury. This would then add to the spectrum of inflammatory organ injury described in critically ill patients (see Table 17.1). This topic is revisited in the very last section of the chapter (A FINAL WORD).

Clinical Features

Both CIP and CIM produce a flaccid paralysis in limb and truncal muscles, associated with hyporeflexia or areflexia (36). However as just mentioned, these conditions often go undetected until there is an unexplained failure to remove a patient from mechanical ventilation.

. The diagnosis of CIP is secured by nerve conduction studies, which show slowed conduction in sensory and motor fibers (41).

. The diagnosis of CIM can be confirmed by electromyography (which shows myopathic change) and by muscle biopsy (which shows loss of myosin filaments, and inflammatory infiltration) (39).

Management

There is no treatment for CIP or CIM. Several measures have been proposed to reduce the risk of

these disorders (e.g., early ambulation, improved nutrition), but none have been validated (and none are directed at the inflammatory nature of the disorders).

Outcomes

These conditions can be devastating in ventilator-dependent patients, because they prevent weaning from ventilatory support. Complete recovery is expected in about half the patients (41), but it can takes months to recover.

DRUG-INDUCED PARALYSIS

Neuromuscular blocking agents produce a flaccid paralysis that is used in the following situations: (*a*) to facilitate endotracheal intubation, (*b*) to prevent shivering during induced hypothermia, and (*c*) to promote synchronous mechanical ventilation in patients who are severely agitated (42).

These agents act by binding to acetylcholine (Ach) receptors on the postsynaptic side of neuromuscular junctions. Once bound, there are two different modes of action: (*a*) *depolarizing agents* produce a sustained depolarization of the postsynaptic membrane, and (*b*) *non-depolarizing agents* act by competitive inhibition of Ach binding to the postsynaptic receptors. Both actions effectively block the transmission of electrical impulses from nerve to muscle.

Commonly Used Agents

The comparative features of three commonly used neuromuscular blocking agents are shown in Table 46.7 (43). *Note:* Many experts recommend that ideal body weight be used for dosing of neuromuscular blockers (42).

TABLE 46.7 Properties of Commonly-Used Neuromuscular Blocking Agents

	Succinylcholine	Rocuronium	Cisatracurium
Intubating Dose (IV)	1–1.5 mg/kg	1.0 mg/kg	0.15–0.2 mg/kg
Onset Time	1–1.5 min	1.5–3 min	5–7 min
Clinical Duration	7–12 min	50–70 min	35–50 min
Infusion Dose	—	5–12 µg/kg/min	1–3 µg/kg/min
Cardiovascular Effects	Bradycardia	None	None
Contraindications	Multiple†	None	None
Influence of Renal or Hepatic Dysfunction	None	Prolonged effect with liver failure	None

†Contraindications to succinylcholine include hyperkalemia, malignant hyperthermia, rhabdomyolysis, burns, and immobility from spinal chord injury.

Properties of the neuromuscular blockers from Reference 43.

Succinylcholine

Succinylcholine is the only depolarizing agent available, and has both a rapid onset of action (60–90 seconds) and a rapid recovery time (7–12 minutes). Because of these features, succinylcholine is popular for rapid-sequence endotracheal intubation.

SIDE EFFECTS: Succinylcholine-induced depolarization of skeletal muscle promotes the efflux of K^+ from muscle cells. This can be associated with up to a 0.5 mEq/L rise in the serum K^+ (44), but this effect is without consequence if the baseline serum K^+ is not elevated. However, life-threatening hyperkalemia can occur when succinylcholine is given to patients with hyperkalemia, or to patients with malignant hyperthermia, rhabdomyolysis, burns, or immobility from spinal cord injury. As a result, succinylcholine is contraindicated in these conditions.

Rocuronium

Rocuronium is a non-depolarizing neuromuscular blocker with a rapid onset of action (1.5–3 minutes) and an "intermediate" duration of clinical effect (30–70 min). It has no cardiovascular side effects, and has gradually replaced vecuronium, a related drug with a more delayed onset of action. Rocuronium has a prolonged effect in patients with hepatic dysfunction (43), but the extent of the prolongation has not been adequately studied.

INTUBATION: Because of the rapid onset of action, rocuronium can be used for endotracheal intubation when succinylcholine is contraindicated. At a dose of 1.0 mg/kg, rocuronium has a rapid onset that is equivalent to a 1.0 mg/kg dose of succinylcholine (45). The only concern with rocuronium is the prolonged duration of effect (about one hour), which can be problematic if the airway is not secured in a timely fashion. In this situation, rapid reversal of rocuronium with sugammadex (16 mg/kg) is advised. (See later for reversal agents.)

Cisatracurium

Cisatracurium is a non-depolarizing agent with a prolonged onset of action (5–7 min) and an intermediate duration of effect (35–50 min). Unlike a related agent (atracurium), cisatracurium does not cause histamine release, and it has no cardiovascular side effects (43). Blood levels of cisatracurium are not influenced by renal or liver failure, which makes it an appealing neuromuscular blocker for ICU patients.

Monitoring

The standard method of monitoring drug-induced paralysis is to apply a series of four low-frequency (2 Hz) electrical pulses to the ulnar nerve at the forearm, and observe for adduction of the thumb. Total absence of thumb adduction is evidence of excessive block. The desired goal is 1 or 2 perceptible twitches, and the drug infusion is adjusted to achieve that end-point (30).

Reversal Agents

There are two agents that can reverse neuromuscular blockade: *neostigmine* and *sugammadex*.

Neostigmine

Neostigmine blocks the breakdown of Ach (by inhibiting the acetylcholinesterase enzyme), and the subsequent increase in Ach is enough to overcome the competitive blockade by the

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nondepolarizing agents. (There are two other "anticholinesterases", edrophonium and pyridostigmine, but neither is used as a reversal agent.)

DOSAGE: The dose of neostigmine needed to reverse neuromuscular paralysis is dependent on several factors, including the neuromuscular blocker, the depth of paralysis, and the anesthetic agent that is used. An intravenous neostigmine dose of 70 μg/kg can reverse a moderate level of paralysis, but the recovery time can vary from 8 to 45 minutes, depending on the other factors just mentioned (43). The dose of neostigmine should never exceed 70 μg/kg because of the risk of parasympathomimetic side effects (43).

Sugammadex

Sugammadex is a reversal agent that binds directly to the aminosteroid class of neuromuscular blockers (i.e., rocuronium, vecuronium, and pancuronium). It binds most tightly to rocuronium, but is equally effective for vecuronium. (The binding to pancuronium is far less extensive.)

DOSAGE: The dose of sugammadex depends on the depth of paralysis. To reverse rocuronium within minutes after it is administered (i.e., after an intubation), the sugammadex dose is 16 mg/kg, and the reversal time is only 2–3 minutes (which is less than the recovery time for succinylcholine) (43). With deep paralysis, the dose of sugammadex is 4 mg/kg; with moderate paralysis, the dose is 2 mg/kg, and the time for reversal is always 2–3 minutes (43). *The reversal with sugammadex is much faster than with neostigmine, and there are no cholinergic side effects*.

A Plea to Avoid Prolonged Paralysis

The experience of being awake while paralyzed is both horrifying and painful (46), and it is imperative to use heavy sedation, and at least some analgesia, while patients are paralyzed. However, it is not possible to evaluate the adequacy of sedation or pain control while a patient is paralyzed without continuous EEG monitoring (which is often not available for this purpose). The inability to ensure adequate sedation and pain control is a major reason to avoid prolonged periods of neuromuscular paralysis whenever possible. This will also reduce the risk of the following complications: (*a*) critical illness myopathy, (*b*) "hypostatic" pneumonia (from pooling of respiratory secretions in dependent lung regions), (*c*) venous thromboembolism, and (*d*) pressure ulcers on the skin.

A FINAL WORD

Inflammation Strikes Again

One of the central themes in this book is the harm inflicted by severe or progressive systemic inflammation in critically ill patients. Inflammatory injury is responsible for the acute respiratory distress syndrome (see Chapter 24) and for many cases of acute kidney injury (see Chapter 34), and inflammation has a pivotal role in the multiorgan failure associated with septic shock (see Chapter 17) and with the post-cardiac arrest period (see Chapter 21). This chapter adds to this list by revealing that inflammation can damage peripheral nerves (i.e., critical illness polyneuropathy) and skeletal muscle (i.e., critical illness myopathy). Considering the swath of

damage that can be inflicted by inflammation, it seems apparent that inflammation is the most lethal force that you (and the patients) will face in the ICU.

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Chapter 47

Acute Stroke in the ICU

Disease makes men more physical, it leaves them nothing but body.

Thomas Mann (*a*)

The focus of this chapter is a cerebrovascular disorder that produces a variety of acute (and often devastating) neurologic deficits. It was first described over 2,400 years ago, and has suffered through a series of inappropriate names, including *acute brain suffering*, *apoplexy*, *stroke*, *cerebrovascular accident* (what accident?), and the most recent embarrassment, *brain attack*. Considering that this disorder is the second leading cause of death worldwide (1), it deserves a better name.

This chapter describes the management of ischemic strokes, as well as strokes caused by spontaneous intracerebral hemorrhage, and subarachnoid hemorrhage. The focus is on management in the ICU, and not on decisions or treatments that occur when the stroke is first suspected (e.g., thrombolytic therapy), which is the purview of neurologists and emergency medicine specialists. Many of the recommendations in this chapter are based on the most recent clinical practice guidelines for each type of stroke (2–5).

ISCHEMIC STROKE

Ischemic strokes account for about 85% of all strokes. Most are the result of atherosclerotic plaques that rupture and trigger a thrombotic occlusion (similar to the pathogenesis of myocardial infarctions), but about 20% are *embolic strokes*. Most emboli are from the left side of the heart, but a small fraction of emboli begin as deep vein thrombosis in the legs, and reach the brain through a patent foramen ovale (see later).

This section describes the management of ischemic strokes after admission to the ICU, and includes general support measures, and special considerations in patients who have received thrombolytic therapy or mechanical thrombectomy. The general support measures that deserve attention are summarized in Table 47.1.

Respiratory Support

Decisions about respiratory support include both oxygenation and ventilation.

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Oxygen

Oxygen has two effects that can be deleterious in acute stroke. The first is oxygen's action as a vasoconstrictor in all major vascular beds except the lungs (where it acts as a vasodilator). For the cerebral circulation, inhalation of 100% O_2 results in a 15–30% decrease in cerebral blood flow (6). The second deleterious effect is the ability of oxygen to promote oxidative cell injury via the production of *reactive oxygen species* (7). These two actions would explain the study showing that maintaining an arterial O_2 saturation (SaO_2) above 94% has a negative impact on neurologic outcomes in patients with acute ischemic stroke (8).

RECOMMENDATION: The recent guidelines on stroke management recommend supplemental O_2 only in patients who are hypoxemic (2). However, the same guidelines also state that the SaO_2 should be kept above 94% (2), which is above the hypoxemic level (i.e., <90%), and is at the same level that was associated with worse neurologic outcomes in the study mentioned previously (8). Therefore, it seems wise to *maintain the SaO_2 at 90-92%*, and no higher.

Mechanical Ventilation

Intubation and mechanical ventilation are indicated for: (*a*) patients with impaired consciousness who are unable to protect their airways from aspiration of mouth or gastric secretions, and (*b*) patients with acute respiratory failure (hypoxemic and/or hypercapnic) who cannot be managed otherwise. Remember that *positive pressure ventilation can reduce cerebral perfusion pressure* by increasing intracranial pressure (see Equation (47.1) later in the chapter), so the judicious use of positive pressure ventilation is wise in patients with ischemic stroke.

TABLE 47.1 General Support Measures for Acute Ischemic Stroke

Concern	Recommendations
Oxygen	• Use supplemental O_2 only for patients who are hypoxemic (SaO_2 <90%).
Hypertension	• After thrombolytic Rx or mechanical thrombectomy, keep BP <180/105 for 24 hrs. • Otherwise, lower BP only if >220/120 for the first 72 hrs after admission (unless there is heart failure, etc).
Hyperglycemia	• Lower the blood glucose if >180 mg/dL, and maintain the level at 140–180 mg/dL for 24–48 hrs after admission. • Avoid dextrose-containing IV fluids.
Fever	• Aggressively treat fever for the first 24–48 hrs. • Search for infection as source of fever.
Antiplatelet Therapy	• Start aspirin, 325 mg daily, on day 1 (to prevent early recurrence). • Following thrombolytic Rx, wait 24 hrs before starting aspirin.
Thrombo-prophylaxis	• All patients should receive prophylaxis for deep vein thrombosis using low-molecular-weight heparin. • Following thrombolytic therapy, use sequential compression devices for the first 24 hours.

Hypertension Management

Elevated blood pressure is reported in over 60% of patients with acute stroke (9), and is

considered to be a physiological response that helps to maintain flow in areas surrounding the infarcted area (i.e., the penumbra), where flow is threatened. This is considered to be especially important because cerebral autoregulation is impaired in patients with acute stroke (10), so cerebral flow is highly dependent on blood pressure. This might explain why *blood pressure reduction in the early period following acute stroke has resulted in no apparent improvement in outcomes* (11).

Recommendations

The following recommendations are from the most recent guidelines on stroke management (2).

. For patients who did not receive thrombolytic therapy, blood pressure reduction is advised only at pressures >220/120 mm Hg for the first 72 hours after the stroke (unless there is a condition like left heart failure that requires a more aggressive approach). Furthermore, any decrease in blood pressure should probably not exceed 15% in the first 24 hours after the stroke.

. For patients who have received thrombolytic therapy, the BP should be <180/ 105 mm Hg for 24 hours.

. There is little agreement on the optimal blood pressure following mechanical thrombectomy with successful reperfusion. The guidelines recommend a BP <180/105 mm Hg for 24 hours (2), while other protocols aim for a normal BP (<140/90) after revascularization (12).

TABLE 47.2 Antihypertensive Treatment in Acute Ischemic Stroke

Drug	Dosage and Comments
Labetalol	Dosage: 10 mg as IV bolus, then infuse at 2–8 mg/min; titrate to desired BP. Comment: Equivalent to nicardipine in efficacy and safety.
Nicardipine	Dosage: Infuse at 5 mg/hr, and increase in increments of 2.5 mg/hr, if needed, to 15 mg/hr. Comment: Equivalent to labetalol in efficacy and safety.
Clevidipine	Dosage: Infuse at 1–2 mg/hr, and double the dose rate every 2–5 min, if needed, to 21 mg/hr. Comment: Equivalent to nicardipine in efficacy and safety.

Drug doses from the guidelines in Reference 2. Drug equivalencies from References 12,13.

Antihypertensive Drugs

The recommended drug regimens for blood pressure reduction in the early post-stroke period are included in Table 47.2. *Labetalol* (a combined α- and β-blocker) and *nicardipine* (a calcium channel blocker) have proven to be equivalent for BP control in acute stroke (12). *Clevidipine* (a calcium channel blocker) is not as popular as the other two agents, but has been shown to be equivalent to nicardipine in patients with cerebrovascular disease (13).

Pure β-blockers are not recommended in acute ischemic stroke because they decrease cardiac output, however *metoprolol* (5–10 mg IV every 6 hours) can be used after mechanical thrombectomy with successful reperfusion (2). Nitroprusside should be avoided in acute stroke because it increases intracranial pressure.

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Hypotension

Considering that cerebral autoregulation is impaired in acute stroke, prompt correction of hypotension is mandatory, using fluids for hypovolemia and vasopressors, if needed.

. The guidelines for stroke management recommend isotonic saline for volume resuscitation (2), based on studies showing no difference in outcomes between colloid and crystalloid fluids in acute stroke (14). However, the tendency of crystalloid fluids to promote edema formation (see Figure 10.1) seems reason enough to use some colloid fluids (i.e., 5% albumin) for volume resuscitation in acute stroke, especially in patients with hypoalbuminemia.

. For vasopressors, phenylephrine is appropriate for patients with rapid atrial fibrillation, otherwise norepinephrine is an appropriate choice.

. There is no recommendation concerning the target blood pressure in acute stroke, but considering that cerebral autoregulation is impaired, a target pressure that is higher than usual (i.e., greater than a mean arterial pressure of 65 mm Hg) seems warranted.

Glycemic Control

Hyperglycemia occurs in 30–50% of patients (diabetics and nondiabetics) in the first 24 hours after acute stroke, and it has a negative impact on neurologic recovery (15,16). This is very similar to the adverse impact of hyperglycemia on neurologic recovery after cardiac arrest (see Chapter 21). Proposed mechanisms include: (*a*) increased production of lactic acid with local acidosis, (*b*) oxidative stress from glucose-initiated production of reactive oxygen species, and (*c*) disruption of the blood-brain barrier with edema in the penumbra region (15).

The guidelines on stroke management recommend treating hyperglycemia if the blood glucose exceeds 180 mg/dL, with *a target blood glucose in the 140–180 mg/dL rang*e (2). (Since the hyperglycemia is transient, tighter glycemic control runs the risk of hypoglycemia.) *Avoiding dextrose-containing IV solutions is also advised*.

Fever

Fever is reported in up to 60% of patients with acute ischemic stroke (16), and the presence of fever in the first 24–48 hrs a negative influence on both neurologic recovery and mortality (17–19). This is similar to the negative impact of fever on neurologic recovery after cardiac arrest (which is the basis of targeted temperature management, described in Chapter 21).

Source of Fever

Fever typically appears on presentation, or within 72 hours after stroke onset (17), which suggests a noninfectious origin (e.g., from tissue necrosis or intracerebral blood). However, some studies have found infections in a majority of patients with stroke-related fever (20), so a search for infection is warranted for all stroke-related fevers.

Antipyretic Therapy

Prompt decrease in body temperature is indicated for fever in the first 24–48 hours after stroke. The dosing of acetaminophen for antipyresis is described in Chapter 44. *Targeted temperature management does not improve outcomes after ischemic stroke* (21).

Other Measures

Thromboprophylaxis

Acute pulmonary embolism is traditionally cited as a common complication of acute stroke (22), although a 12-year survey of 3 million patients with acute stroke (in Germany) showed only a 0.4% incidence of PE (23). Nevertheless, prophylaxis for deep vein thrombosis is warranted, and low-molecular-weight heparin is the appropriate choice (see Table 5.2 for dosing recommendations). For patients who have received thrombolytic therapy, intermittent pneumatic compression can be used for the first 24 hours.

Antiplatelet Therapy

Aspirin, 325 mg daily, should be started on admission to the ICU (or on day 2 if the patient has received thrombolytic therapy). The aim is to prevent early recurrent stroke, and estimates are that one stroke is avoided for every 100 patients treated (24). Aspirin can be given either orally or by rectal suppository. Anticoagulant therapy is not recommended for acute stroke.

Echocardiography

The following are indications for a cardiac ultrasound in patients with acute ischemic stroke:

. Embolic stroke is suspected (e.g., neuroimaging shows multiple areas of infarction).

. Atrial fibrillation (past or present).

. History of transmural MI (for possible mural thrombus).

. Positive blood cultures (for possible endocarditis).

. History of venous thromboembolism (for possible patent foramen ovale)

. Cryptogenic stroke (same as #5).

Transesophageal ultrasound is the most sensitive method for detecting valvular vegetations or thrombi in the cardiac chambers, but it can be reserved for instances where transthoracic ultrasound (which is much easier to perform) is unrevealing.

Patent Foramen Ovale

As mentioned earlier, an embolic stroke can be the result of a thrombus that originates in the proximal leg veins and passes through a patent foramen ovale (PFO) to embolize in the cerebral circulation. Although uncommon, this scenario is a consideration when there is no apparent predisposing condition for ischemic stroke (e.g., in patients younger than 55 years of age) (25), or in patients who have a history of venous thromboembolism. The test described next is a reliable method for the detection of PFOs.

BUBBLE TEST: This test can be performed with transesophageal or transthoracic echocardiography. For the transthoracic approach, an apical 4-chamber view is preferred. About 10 mL of saline is first agitated to create microbubbles (i.e., by rapidly filling and emptying two syringes attached by a 3-way stopcock). The saline is then rapidly injected, either through a central venous catheter, or a peripheral catheter in the upper arm. After injection, the

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microbubbles will become visible in the right side of the heart, and in the presence of a right-to-left intracardiac shunt (from a PFO), the bubbles will appear in the left side of the heart after a few cardiac cycles. This is demonstrated in Figure 47.1 (26).

The presence of a PFO in the setting of an acute ischemic stroke is not evidence of a causal link between the two. However, it is sufficient reason to close the PFO at a later time (if the patient has a satisfactory neurological recovery), using a catheter-based technique.

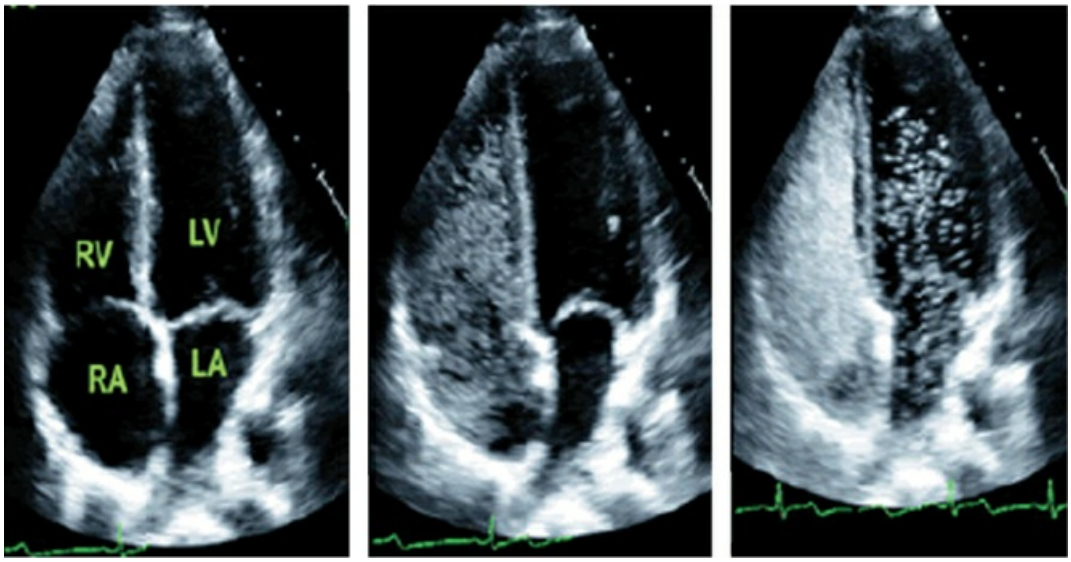


FIGURE 47.1 Transesophageal echocardiography in a patient with acute ischemic stroke, showing microbubbles (from the injection of agitated saline) moving from the right to left side of the heart. This is evidence of a right-to-left shunt from a patent foramen ovale. Images adapted from Reference 26.

CNS Complications

The central nervous system complications of acute stroke, (and stroke management), include intracerebral hemorrhage, cerebral edema, and seizures.

Intracerebral Hemorrhage

The most devastating complication of stroke management is shown in Figure 47.2 (27). The CT images in this figure are from a patient with an acute ischemic stroke who experienced an acute deterioration in mental status five hours after receiving thrombolytic therapy with tissue plasminogen activator (tPA). A repeat CT image at that time showed a massive intracerebral hemorrhage with mass effect.

The incidence of symptomatic intracerebral hemorrhage following thrombolytic therapy for acute stroke is 6.4% to 8.8% (28,29). Risk factors include advanced age, blood pressure above 180/105, and a stroke that involves more than one-third of a cerebral hemisphere. The risk of hemorrhage is the reason that patients who receive thrombolytic therapy are admitted to an ICU or stroke unit for 24 hours, where it is possible to closely monitor neurologic status and blood pressure. Anticoagulants and antiplatelet agents are also withheld for the first 24 hours after thrombolytic therapy.

MANAGEMENT: There is no validated treatment for hemorrhagic transformation of acute stroke,

but the following measures are recommended in the most recent guidelines on stroke management (2).

. Cryoprecipitate (10 Units) is recommended as a source of fibrinogen, and the dose can be repeated if the serum fibrinogen is below 200 mg/dL (2). However, cryoprecipitate must be thawed, which delays treatment by at least 20 minutes. A better choice would be *fibrinogen concentrates*, which have two advantages over cryoprecipitate (30): (*a*) thawing is not necessary, so treatment can be started immediately, and (*b*) the concentration of fibrinogen is more consistent (at 20 g/L).

. If cryoprecipitate is not immediately available, tranexamic acid (an antifibrinolytic agent) can be given in a dose of 1,000 mg IV over 10 minutes (2).

. A neurosurgery consultation is warranted (to consider decompression hemicraniectomy).

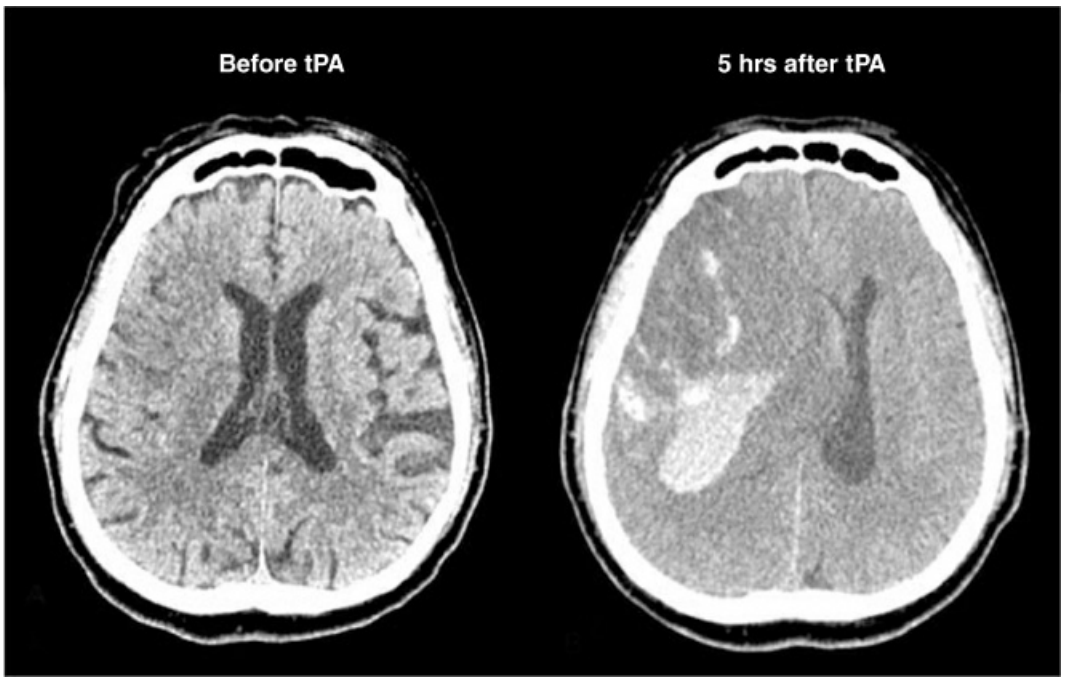


FIGURE 47.2 Noncontrast CT images of the brain from a patient who experienced an abrupt deterioration in mental status 5 hours after receiving thrombolytic therapy with tissue plasminogen activator (tPA). The CT scan on the right shows a massive intracerebral hemorrhage (hyperdense areas) with midline shift. Images from Reference 27.

Cerebral Edema

Hemorrhage and cerebral edema are the principal determinants of poor outcomes after acute stroke. There are two types of cerebral edema: *cytotoxic edema*, which is the result of cell necrosis, and *vasogenic edema*, which involves loss of vascular control mechanisms, and results in disruption of the blood-brain barrier. The larger the ischemic area, the more extensive the cerebral edema.

MANAGEMENT: Intracranial pressure (ICP) monitoring is not recommended for acute ischemic stroke that is complicated by cerebral edema (2). Instead, monitoring neurologic symptoms and

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signs is considered adequate. When cerebral edema causes a deterioration in neurologic status, *hypertonic saline* can be used to lower the ICP (31,32). Mannitol is also capable of lowering the ICP, but it is not as effective as hypertonic saline in patients with an elevated ICP after stroke (33). There is no evidence that hypertonic therapy alters the outcome in acute stroke (31,32).

There are a number of different hypertonic saline solutions, and Table 47.3 shows some of the characteristics of the different solutions, including the bolus doses recommended for lowering the ICP (32). The more concentrated solutions are appealing because they are effective at a lower volume. Bolus dosing of hypertonic saline is preferred to a continuous infusion because the effect occurs more rapidly. The target plasma sodium is typically 145–155 mEq/L (32).

TABLE 47.3 Hypertonic Saline Solutions for Treating Cerebral Edema

	3% NaCL	5% NaCL	7.5% NaCL	14.6% NaCL	23.4% NaCL
$[Na^+]$ (mEq/L)	513	856	1,293	2,500	4,000
Osmolality (mOsm/L)	1,027	1,711	2,566	5,000	8,008
Commercially Available	Yes	Yes	No	Yes	Yes
Bolus Dose for Cerebral Edema	250 mL or 5 mL/kg	100 mL	100 mL or 2–4 mL/kg	24–48 mL	30 mL or 0.68–2 mL/kg

From Reference 32.

Seizures

About 20% of patients have a symptomatic seizure in the first week after an ischemic stroke, and about one-quarter of the seizures recur at some time in the ensuing 4 years (34). Therefore, anticonvulsant therapy (e.g., with levetiracetam) is recommended for post-stroke seizures, and it should be continued after discharge. Seizure prophylaxis is not recommended in the early period following ischemic stroke (2).

SPONTANEOUS INTRACEREBRAL HEMORRHAGE

Spontaneous (non-traumatic) intracerebral hemorrhage accounts for about 10–15% of acute strokes, and the predisposing conditions include hypertension and systemic anticoagulation. The prognosis with this type of stroke is typically less favorable than with ischemic strokes: i.e., about half of the patients do not survive for 30 days (35), and two-thirds of the survivors have significant neurologic deficits (36). The CT image in Figure 47.3 helps to understand why the prognosis is poor following intracerebral hemorrhage.

Reversing Antithrombotic Agents

Rapid deterioration is common in the early hours after intracerebral hemorrhage (ICH), and is attributed to expansion of the hematoma (3). Therefore, the first concern is to arrest growth of the hematoma, and to this end, prompt reversal of ongoing antithrombotic therapy is mandatory. The reversal agents and dosing regimens for the commonly used antithrombotic agents are presented in Tables 47.4 and 47.5 (36–38).

Warfarin

Anticoagulation with warfarin can be reversed by 4-factor prothrombin complex concentrate (PCC) (so named because it contains Factors II, VII, IX, and X), which can normalize the INR much faster than fresh frozen plasma (FFP), and at a much smaller volume. In one study of excessive bleeding from warfarin, 4-factor PCC normalized the INR within 30 minutes in 60% of the patients, while FFP achieved the same feat in only 9% of patients (39). Vitamin K is given along with the 4-factor PCC, but the effect is much slower (i.e., the effect begins at about 2 hours, and is maximal at 24 hours) (3). The dosing of 4-factor PCC is determined by the INR, as shown in Table 47.4.

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FIGURE 47.3 Noncontrast CT image showing a large hyperdense intracerebral hematoma that occupies about one-half of the volume of the right side of the brain. The bleeding also extends into the ventricles, which creates the risk of an obstructive hydrocephalus.

Dabigatran

The anticoagulant effect of the direct thrombin inhibitor dabigatran can be reversed in 10–30 minutes by *idarucizumab* (Praxbind®), a monoclonal antibody that readily binds to dabigatran. (See Table 47.4 for the dosing.) An alternative approach is 4-factor PCC in high doses (50 IU/kg) (3), although there is limited experience with this approach.

Factor Xa Inhibitors

The anticoagulant effects of the Factor Xa inhibitors, apixaban and rivaroxaban, can be blocked by *andexanet alfa*, which is a dummy Factor Xa that binds the drugs but produces no anticoagulation. The dosing of this agent is presented in Table 47.5, and is determined by the anticoagulant and dosage, and the time elapsed from the last dose (37). When andexanet is given to patients with excessive bleeding (including intracerebral bleeding) from apixaban or rivaroxaban, there is a 90% reduction in anti-Xa activity after about 10 minutes, and hemostasis

is fully restored after 12 hours (40). Andexanet has one adverse effect that deserves mention: i.e., about 10% of patients develop venous thromboembolism (40). If andexanet is not available, 4-factor PCC in a high dose (50 IU/kg) is recommended as an alternative treatment (3).

TABLE 47.4 Reversal of Antithrombotic Agents in Intracranial Hemorrhage

Antithrombotics	Reversal Agents & Dosage
Warfarin	• Vitamin K: 10 mg IV, plus • 4-factor PCC, dosing as follows: INR ..... Dose 2–4 ..... 25 U/kg 4–6 ..... 35 U/kg >6 ..... 50 U/kg
Dabigatran (Pradaxa®)	• Idarucizumab 2.5 mg IV bolus and repeat within 15 minutes. • Consider high-dose 4-factor PCC (50 IU/kg) if idarucizumab is not available.
Apixaban (Eliquis®) or Rivaroxaban (Xarelto®)	• Andexanet alfa - see Table 47.5 for dosing. • Consider high-dose 4-factor PCC (50 IU/kg) if andexanet is not available.
Unfractionated Heparin	• Protamine sulfate: 1 mg per 100 units of heparin given at last dose.
Enoxaparin (Lovenox®)	• Protamine sulfate: 1 mg per 1 mg of enoxaparin at last dose. Less effective than against unfractionated heparin.
Aspirin, Clopidogrel	• Desmopressin: 3 µg/kg subQ or IV. Can repeat once.

From References 36,37. PCC = prothrombin complex concentrate.

TABLE 47.5 Dosing Recommendations for Andexanet Alfa

Regimen	Criteria	Dosage
Low-Dose Regimen	Last apixaban dose ≤5 mg or rivaroxaban dose ≤10 mg. or time since last dose ≥8 hrs.	Start with IV bolus of 400 mg (at 30 mg/min), then infuse at 4 mg/min for up to 120 min.
High-Dose Regimen	Last apixaban dose >5 mg or rivaroxaban dose >10 mg. or time since last dose <8 hrs or unknown.	Start with IV bolus of 800 mg (at 30 mg/min), then infuse at 8 mg/min for up to 120 min.

From Reference 37.

Antiplatelet Agents

Aspirin and clopidogrel produce irreversible inhibition of platelet adhesiveness that lasts for the lifetime of the exposed platelets (i.e., 7–10 days), but this effect can be antagonized by desmopressin (41), which promotes the release of von Willebrand factor from endothelial cells.

Other Measures

Many of the measures described for ischemic stroke (e.g., blood pressure control, glycemic control) are also recommended for patients with spontaneous ICH (3).

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. High blood pressure can be especially harmful in acute hemorrhage, and maintaining a systolic blood pressure below 140 mm Hg is recommended (3).

. Prophylaxis for deep vein thrombosis should begin at the time of admission, and sequential compression devices should be used for prophylaxis in the first 24 hours. Thereafter, the decision to begin prophylaxis with one of the heparin preparations will depend on the clinical condition of the patient (stable or unstable).

. Seizure prophylaxis is not recommended in patients with ICH (3). However, symptomatic seizures are reported in about 15% of patients in the first week after ICH (3), and these patients should receive anticonvulsant therapy (e.g., with levetiracetam).

. Patients with ICH who have impaired consciousness should be evaluated for nonconvulsive status epilepticus with continuous EEG monitoring for 24–36 hours, if available.

. A neurosurgery consult is warranted in most cases of ICH, especially those with large or expanding hematomas, or bleeding that has extended into ventricles (as in Figure 47.3). Intraventricular blood often leads to obstructive hydrocephalus, and requires insertion of a catheter into the ventricles for drainage, and for intraventricular thrombolytic therapy (42). The intraventricular catheter can also be used to monitor the intracranial pressure.

Intracranial Pressure

The clinical practice guideline on spontaneous ICH (3) recommends intracranial pressure (ICP) monitoring in the following situations: (*a*) patients with a Glasgow Coma Score of ≤8 (see Table 45.5), (*b*) blood in the ventricular system, and (*c*) clinical evidence of transtentorial herniation (e.g., unilateral pupillary dilation). The ICP should then be maintained below 20 mm Hg (usually by periodically draining fluid from the intraventricular catheter) (3,43). The ICP measurement also allows an estimation of the *cerebral perfusion pressure*.

Cerebral Perfusion Pressure

The cerebral perfusion pressure (CPP) is the driving pressure for flow in the cerebral circulation, and is the difference between the mean arterial pressure (MAP) and the intracranial pressure (ICP); i.e.,

$$CPP = MAP - ICP \text{ (mm Hg)} \qquad (47.1)$$

The influence of CPP on cerebral blood flow is influenced by the vascular resistance in the cerebral circulation, and the presence or absence of cerebral autoregulation. *A CPP of 50–70 mm Hg is considered a reasonable goal in patients with ICH* (44).

SUBARACHNOID HEMORRHAGE

The most devastating type of stroke occurs when a saccular aneurysm in one of the cerebral arteries ruptures, resulting in the accumulation of blood in the subarachnoid space (see Figure 47.4) (45). The severity of this condition is demonstrated by the following observations:

. For patients who experience an aneurysmal SAH, as many as 25% will succumb before

reaching the hospital (4), and another 20% will not survive the hospitalization (46).

. Six months after hospital admission, 35% of the patients will be dead or disabled (46).

. Combining these observations indicates that over half of the patients with an aneurysmal SAH will die or become disabled within 6 months.

SAH also affects a younger age group than the other types of stroke (4). In one survey, the mean age of SAH victims was 55 years (47). Thus, SAH can remove people from their source of income, which has serious implications for families.



FIGURE 47.4 Postmortem specimen showing blood in the subarachnoid space at the base of the brain. The source of the subarachnoid hemorrhage was rupture of a saccular aneurysm in the basilar artery, which is located under the hematoma, as indicated by the arrow. Image from Reference 45.

Sources of Tissue Injury

The poor outcomes in aneurysmal SAH are explained by the multiple sources of tissue injury, as

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summarized below.

. The presence of blood in the subarachnoid space triggers an intense inflammatory response that damages the blood brain barrier (which promotes vasogenic cerebral edema) and produces oxidative damage in neurons and glial cells (which produces cytotoxic cerebral edema) (48). The iron in extravasated blood very likely has role in the damaging effects of subarachnoid hemorrhage, because iron promotes the production of highly-toxic hydroxyl radicals (see Figure 25.6), and free iron can promote cell death by oxidizing membrane lipids; a process known as *ferroptosis* (49).

. Blood in the subarachnoid space blocks the flow of cerebrospinal fluid, which often leads to an obstructive hydrocephalus and increased intracranial pressure.

. In about one-third of patients, there is a second phase of tissue injury that is ischemic in nature, and occurs between days 4 and 14 (4). This *delayed cerebral ischemia* (DCI) is considered a major source of morbidity and mortality, and is attributed to vasospasm and thrombotic occlusion of cerebral arteries and arterioles (50).

. SAH is often associated with inflammatory injury in organs other than the brain. Common associated conditions include the acute respiratory distress syndrome (ARDS), and acute kidney injury (AKI) (4).

Management

The management of SAH is directed at the sources of injury just mentioned. Many of the measures are similar to those described for intracerebral hemorrhage.

Immediate Concerns

The most immediate concern is to reduce the risk of bleeding from the ruptured aneurysm. This requires attention to the following:

. Ongoing antithrombotic therapy should be reversed as soon as possible (see Tables 47.4 and 47.5).

. Correction of hypertension is warranted, if necessary. The clinical practice guidelines for SAH do not recommend a target blood pressure, but there is evidence that rebleeding is uncommon if the systolic BP is below 140 mm Hg (51). Therefore, a systolic BP of ≤140 mm Hg is a reasonable target. Blood pressure reduction should not increase the cardiac output (since this could promote bleeding), so labetalol is an appropriate antihypertensive agent (See Table 47.2 for dosing instructions).

. Antifibrinolytic agents (e.g., tranexamic acid) are not effective in preventing rebleeding (4), and thus are not recommended.

Attention to the Aneurysm

The best way to prevent rebleeding is to obliterate the ruptured aneurysm (e.g., with clips), and you will need a neurosurgeon or interventional radiologist for this task. This may be difficult to pursue if the patient is unstable, but outcomes are improved if corrective action is taken within 3 days of the event (52), and you are advised not to delay this for longer than 7–10 days (4).

Delayed Cerebral Ischemia

The delayed cerebral ischemia (DCI) mentioned earlier, which can appear 4 to 14 days after SAH, is considered a major source of morbidity and mortality. The following measures are recommended to reduce the risk of DCI (4).

. The calcium channel blocker *nimodipine* is recommended orally in a dose of 60 mg every 4 hours, starting as soon as possible after the diagnosis of SAH, and continuing for 3 weeks (4). This dosing regimen has been shown to reduce neurologic deficits, but has no impact on survival (53). The intravenous route is not recommended because of the risk of hypotension.

. Monitoring to detect the onset of DCI can be carried out in several ways. Clinical signs of DCI can include a new focal deficit, or a drop of 2 points in the Glasgow Coma Score (see Table 45.5), so frequent neurologic assessments are advised. CT angiography (CTA) and transcranial Doppler ultrasound (measures flow velocity usually in the middle cerebral artery) are both useful, with sensitivities of about 90% for the detection of cerebral vasospasm (4).

. For severe cases of vasospasm, intra-arterial vasodilators or angioplasty should be considered.

Intracranial Pressure

Recommendations for intracranial pressure (ICP) monitoring are the same as described for intracerebral hemorrhage. Drainage catheters are placed in patients who develop hydrocephalus, and these catheters can be used to monitor the ICP. In general, the ICP is kept below 20 mm Hg with periodic drainage of cerebrospinal fluid through the catheter. Patients who survive will require permanent shunting of cerebrospinal fluid into the abdominal cavity.

General Measures

General support is similar to that described for the other types of stroke, and includes the following:

. The incidence of venous thromboembolism in SAH is as high as 25% (4), so thromboprophylaxis is mandatory. Prophylactic doses of enoxaparin (40 mg subQ daily) have been used safely after aneurysm repair (4).

. Fever is common after SAH, and it has a negative influence on outcomes (54). However, fever suppression with antipyretic agents and cooling devices have not improved outcomes (4).

. Hyperglycemia in the first 72 hours after SAH has a negative impact on outcomes (4), but there is no recommendation about glycemic control in the SAH guidelines (4).

. Hyponatremia is reported in 35% of patients with SAH (55), and is possibly a result of cerebral salt wasting. However, the clinical significance of this has yet to be determined.

A FINAL WORD

Where's the Beef?

The ability of thrombolytic therapy and mechanical thrombectomy to re-establish flow has created high expectations for the management of ischemic strokes, and these expectations fueled

a massive effort to create “stroke centers” in major hospitals, with specialized “stroke teams” to direct the management of acute stroke. The following is an accounting of what this effort has accomplished. (The following numbers are from References 56–60.)

Number of strokes each year in the United States. 795,000
Number of ischemic strokes (85%). 676,000
Number of stroke patients who receive lytic therapy (5%) and/or
mechanical thrombectomy (2%): i.e., 7% of 696,000. 47,320
Number of patients who benefit from reperfusion therapy (1 in 9). 5,258
Percent of strokes that benefit from reperfusion therapy (5,258/676,000). . . 0.8%
Enough said.

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